

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME : CSA0716 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

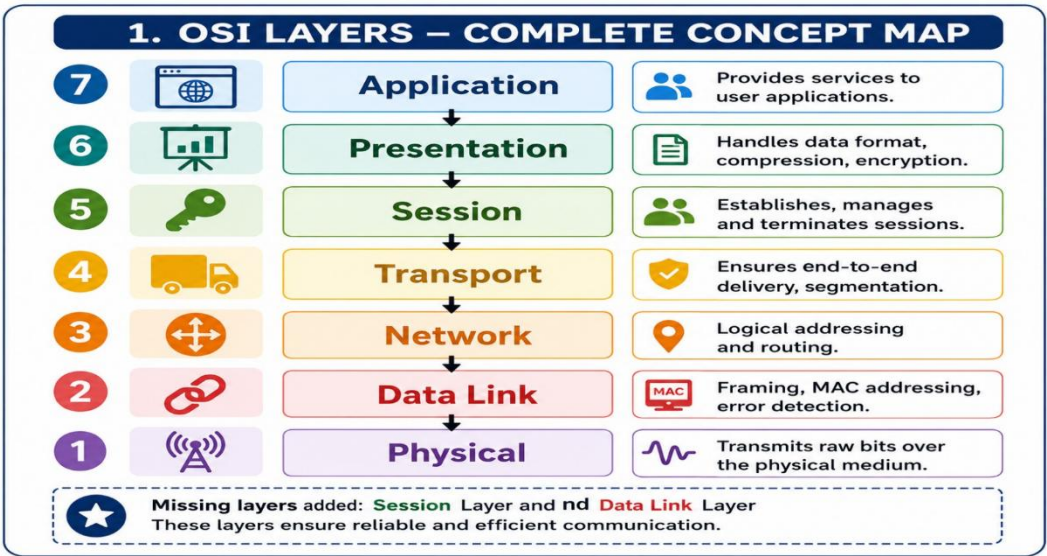
Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.



The concept map represents the OSI (Open Systems Interconnection) Model, which has 7 layers:
Application → Presentation → Session → Transport → Network → Data Link → Physical
So, the missing layers are:

- 1. **Session Layer** – between Presentation and Transport
- 2. **Data Link Layer** – between Network and Physical

Functions of Missing Layers:

1. Session Layer

- Establishes, manages, and terminates communication sessions between devices.
- Controls the dialogue between applications.
- Provides synchronization and recovery during communication.

2. Data Link Layer

- Provides reliable **node-to-node data transfer**.
- Divides data into **frames**.
- Performs **error detection** and controls access to the transmission medium.
- Uses **MAC addresses** for communication within a network.

Importance for Reliable Communication:

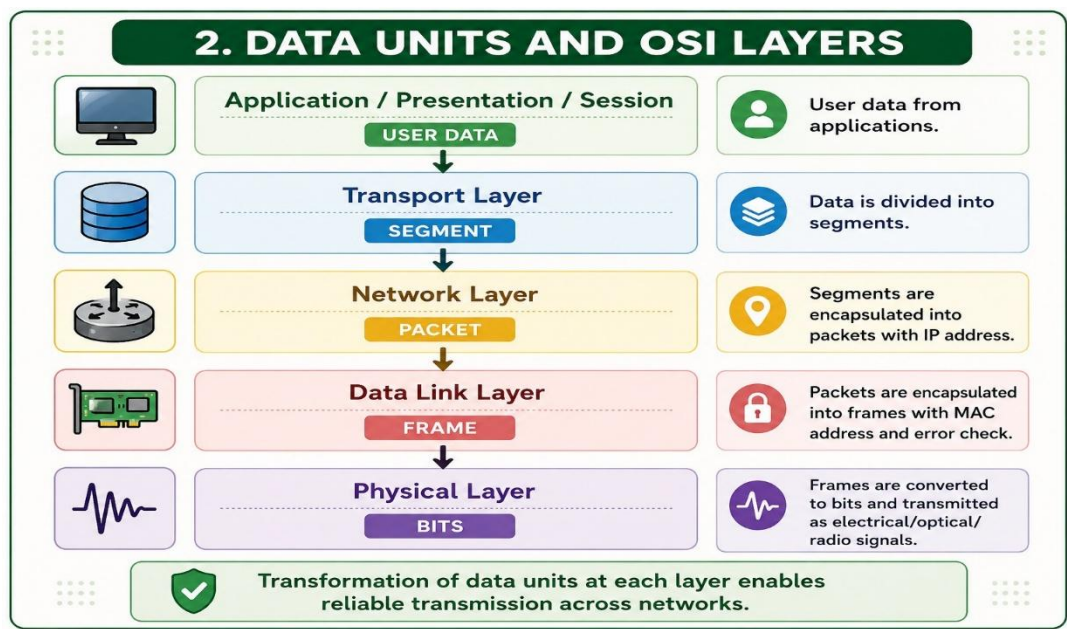
The **Session Layer** ensures that communication between applications is properly maintained and synchronized, while the **Data Link Layer** ensures that data is transferred reliably between directly connected devices by detecting errors and organizing data into frames.

Therefore, the complete OSI concept map is:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Together, these layers ensure **efficient, organized, and reliable end-to-end communication** across a network.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.



Analysis of Data Transformation:

1. Transport Layer – Segment:

The data is divided into **segments**. Port numbers and sequencing information are added to

support reliable end-to-end delivery.

2. **Network Layer – Packet:**

The segment is encapsulated into a **packet**. Logical addresses such as **IP addresses** are added to determine the source and destination.

3. **Data Link Layer – Frame:**

The packet is placed inside a **frame**. MAC addresses and error-detection information are added for reliable delivery between directly connected devices.

4. **Physical Layer – Bits:**

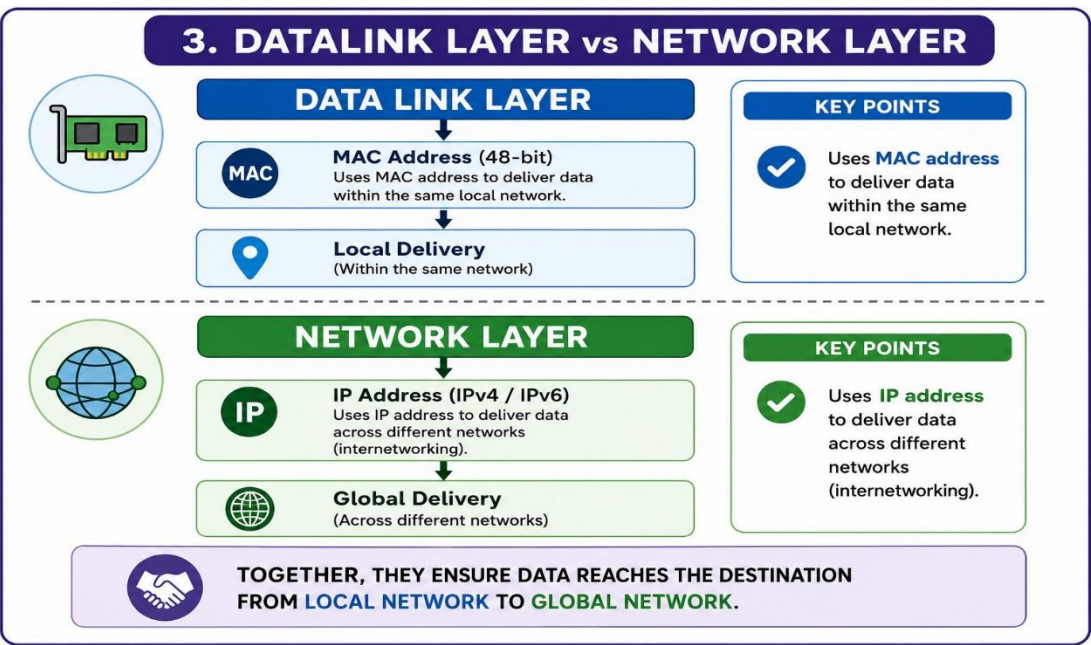
The frame is converted into a stream of **0s and 1s (bits)** and transmitted through cables, fiber, or wireless signals

3. Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.



Step-by-Step Explanation:

1. **Data Link Layer → MAC Address**

- The **Data Link Layer** uses **MAC addresses** to identify devices within the same local network.
- A MAC address uniquely identifies a network interface.

2. **MAC Address → Local Delivery**

- MAC addresses are used to deliver **frames** from one device to another within the **same Local Area Network (LAN)**.
- Switches use MAC addresses to forward frames to the correct device.

3. **Network Layer → IP Address**

- When the destination is on another network, the **Network Layer** uses **IP addresses** to identify the source and destination networks.
- Routers examine IP addresses to determine the best path for the packet.

4. **IP Address → Global Delivery**

- IP addresses enable packets to travel between **different networks** through routers.
- This allows data to reach devices anywhere on interconnected networks, such as the Internet.

Overall Relationship:

Data → IP Address (Global Routing) → Router → MAC Address (Local Delivery) → Destination

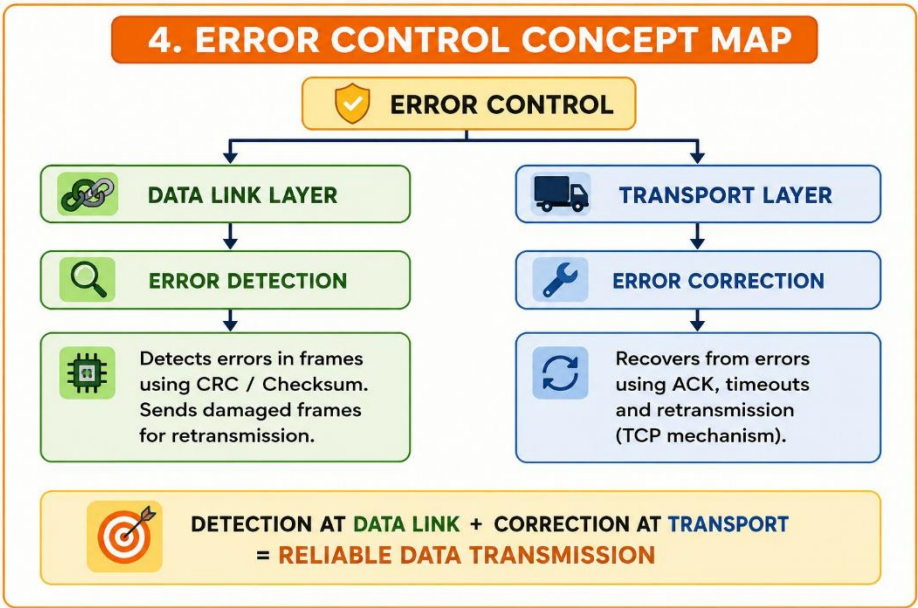
4. A concept map shows:

Error Control

→ **Datalink Layer → Error Detection**

→ **Transport Layer → Error Correction**

Analyze how the interaction between these layers improves reliability in data transmission.



Step-by-Step Explanation:

1. Error Control

- Error control is used to identify and handle errors that may occur during data transmission.
- It helps ensure that the data received is accurate and complete.

2. Data Link Layer → Error Detection

- The Data Link Layer detects errors that occur during **node-to-node communication**.
- It uses techniques such as **CRC (Cyclic Redundancy Check)** to check whether a frame has been corrupted.
- If an error is detected, the damaged frame can be discarded or retransmitted.

3. Transport Layer → Error Correction / Recovery

- The Transport Layer provides **end-to-end reliability**.
- Protocols such as **TCP** use acknowledgements, sequence numbers, and retransmission to ensure that missing or corrupted data is recovered.
- It ensures that data arrives at the destination **in the correct order and without loss**.

Interaction Between Layers:

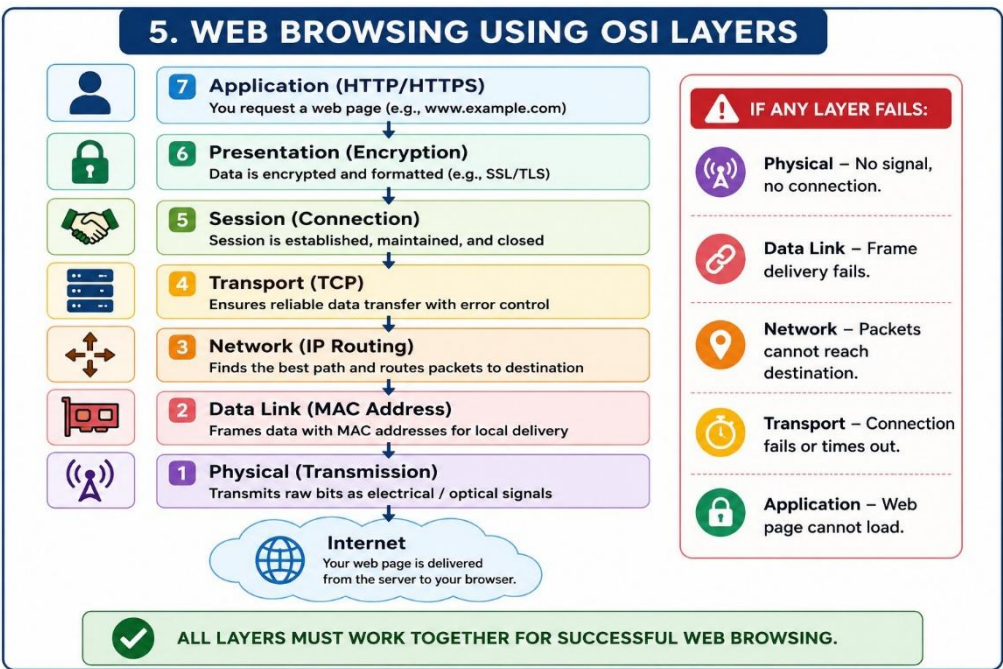
Data Link Layer

- Detects errors during **local transmission**
- Prevents corrupted frames from moving forward

Transport Layer

- Provides **end-to-end error recovery**
- Retransmits lost or missing data
- Ensures correct order and completeness

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.



Analysis of Layer Failure:

OSI Layer	Failure	Effect on Web Browsing
Application	HTTP/DNS problem	Website cannot be requested or found
Presentation	Encryption/formatting failure	Secure website may not load correctly
Session	Session interruption	Communication session may terminate
Transport	TCP connection failure	Data cannot be reliably transferred
Network	IP/routing failure	Packets cannot reach the web server
Data Link	MAC/frame error	Data cannot move correctly through the local network
Physical	Cable/Wi-Fi failure	No physical connection; communication stops

Overall Analysis:

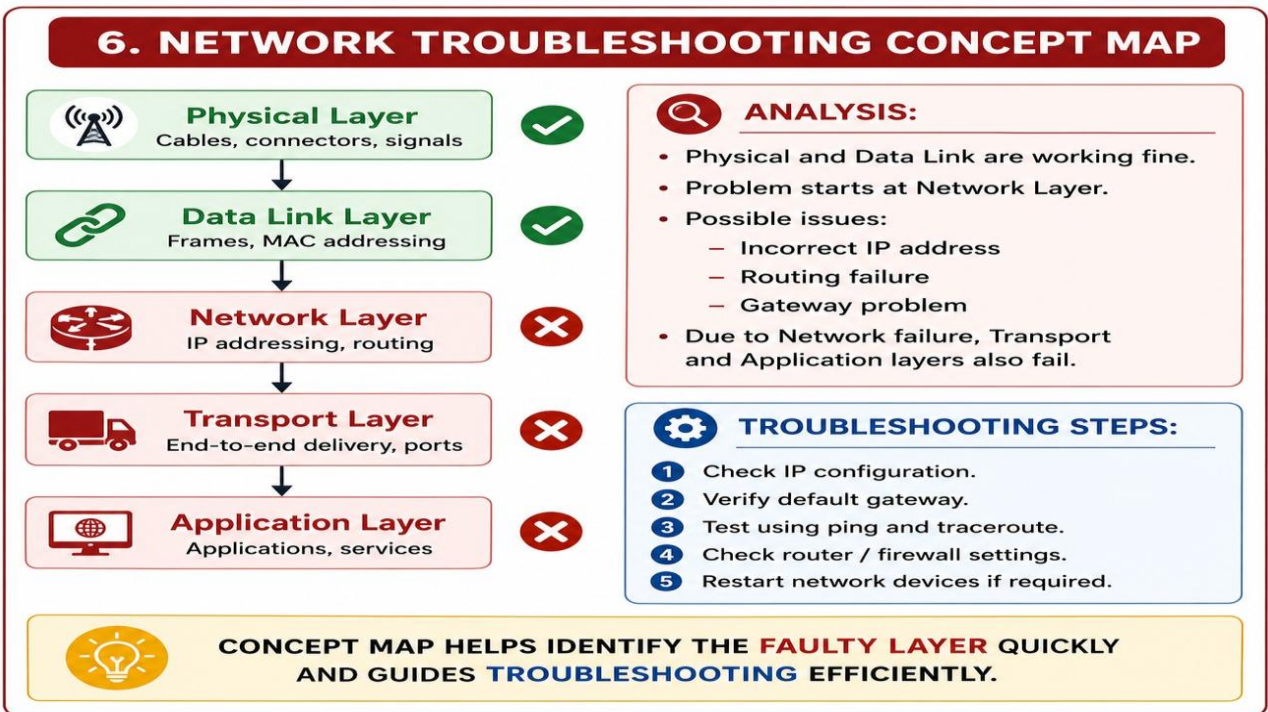
Web browsing depends on the **cooperation of all seven OSI layers**. Data moves **down the OSI layers** at the sender, travels through the network, and moves **up the layers** at the receiver. Each layer performs a specific function, such as formatting, encryption, reliable delivery, routing, framing, and signal transmission.

Failure in any one layer can interrupt or prevent the entire web browsing process. For example, even if the browser and TCP connection work correctly, a failure at the **Physical Layer** (such as no Wi-Fi connection) prevents data from being transmitted. Therefore, the OSI layers work together as a complete system to ensure successful web communication.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.



Problem Analysis:

1. Physical Layer

- The physical connection is working properly.
- Cables, Wi-Fi signals, and network hardware are functioning.

2. Data Link Layer

- Local network communication is working.
- MAC addressing and frame delivery are functioning correctly.

3. Network Layer — Main Problem

- The problem is likely related to IP addressing, routing, or router configuration.
- The device may have an incorrect IP address, subnet mask, default gateway, or routing issue.
- Packets cannot reach the intended destination network.

4. Transport Layer

- Since the Network Layer cannot deliver packets, a reliable TCP connection cannot be established.
- Services using TCP may fail.

5. Application Layer

- Applications such as web browsers, email, or other network services cannot communicate with remote servers.
- This is a result of the lower-layer failure, not necessarily the root cause.

How the Concept Map Helps in Troubleshooting:

The concept map provides a layer-by-layer troubleshooting approach. Since Physical and Data Link are already working, we can skip checking cables, Wi-Fi signals, switches, and MAC addresses.

The troubleshooting should focus on the Network Layer first:

Check IP Configuration → Check Default Gateway → Test Router Connectivity → Check Routing
→ Check DNS → Test Application

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation